Worksheet 7A: Introduction to Matrices — Solutions

Series 7: Linear Algebra
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

Instructor Copy — Not for Distribution

Note to instructor: Key answers are given below. Accept any mathematically equivalent form unless the question specifies otherwise.

Part A

1. 3×3
2. (a) 4 (b) 6 (c) 5 (d) 8
3. $\begin{bmatrix} 1 & 0 & 6 \end{bmatrix}$
4. $\begin{bmatrix} -2 \\ 6 \\ 8 \end{bmatrix}$
5. Any 4×3 matrix with 4 rows (examples) and 3 columns (features).
6. $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
7. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
8. (a) Square: 2×2, 4×4. (b) e.g. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Part B

9. $\begin{bmatrix} 4 & 4 \\ -1 & 6 \end{bmatrix}$
10. $\begin{bmatrix} 3 & -3 \\ 1 & 5 \end{bmatrix}$
11. $\begin{bmatrix} 9 & -3 \\ 6 & 12 \end{bmatrix}$
12. $\begin{bmatrix} -1 & 11 \\ -8 & 0 \end{bmatrix}$

13. $\begin{bmatrix} 4 & 6 \\ 0 & 5 \end{bmatrix}$

14. $A + B = B + A = \begin{bmatrix} 4 & 4 \\ -1 & 6 \end{bmatrix}$. Yes, addition is commutative.

15. $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$, dimensions 3×2.

16. $S^T = S$ (check by swapping rows and columns — they are identical).

17. $W_{\text{new}} = \begin{bmatrix} 0.46 & -0.28 \\ 1.14 & 0.70 \end{bmatrix}$

18. Subtracting moves weights in the direction that reduces the loss (gradient points uphill; we go downhill).

19. Both are 3×4.

20. No — addition requires identical dimensions; $2 \times 3 \neq 3 \times 2$.

Part C

21. (i) Defined, 2×4 (ii) Not defined (iii) Defined, 4×3 (iv) Defined, 1×1

22. $AB = \begin{bmatrix} 7 & -2 \\ 19 & -4 \end{bmatrix}$

23. $BA = \begin{bmatrix} 5 & 10 \\ -2 & -2 \end{bmatrix}$. No, $AB \neq BA$: matrix multiplication is *not* commutative in general.

24. $AI = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A$. Multiplying by I leaves a matrix unchanged.

25. $AB = \begin{bmatrix} 2(1) + 0(0) + 1(3) & 2(4) + 0(-2) + 1(1) \\ -1(1) + 3(0) + 2(3) & -1(4) + 3(-2) + 2(1) \end{bmatrix} = \begin{bmatrix} 5 & 9 \\ 5 & -12 \end{bmatrix}$

26. $A\mathbf{x} = \begin{bmatrix} 1(3) + 0(-1) + (-1)(2) \\ 2(3) + 1(-1) + 0(2) \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

27. $W\mathbf{x} = \begin{bmatrix} 0.2 + 1.6 \\ -0.5 + 0.6 \\ 0.1 + 1.2 \end{bmatrix} = \begin{bmatrix} 1.8 \\ 0.1 \\ 1.3 \end{bmatrix}$; $W\mathbf{x} + \mathbf{b} = \begin{bmatrix} 1.9 \\ -0.1 \\ 1.6 \end{bmatrix}$

28. W is 3×2, $\mathbf{x}$ is 2×1, $\mathbf{y}$ is 3×1. Next layer weight matrix: 2×3.

29. (a) $\begin{bmatrix} 3 & -2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \\ 2 \end{bmatrix}$

30. $A\mathbf{x} = \begin{bmatrix} 3(2) + (-2)(1) \\ 1(2) + 4(1) \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$. But $\mathbf{b} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$... check: $x = 2$, $y = 1$: $3(2) - 2(1) = 4 \neq 5$. The solution to (a) is $x = 2$, $y = -\frac{1}{2}$; verify accordingly.

31. $(A+B)C = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 1 \\ 2 & 2 \end{bmatrix}$; $AC+BC$: same result. This is the **distributive law**.
32. (a) False. (b) True. (c) True.
33. $\hat{\mathbf{y}} = X\mathbf{w}$ has shape 100×1. Each entry is the predicted value for one training example.
34. $R\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (up); $R\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ (left). Yes, consistent with 90° anticlockwise rotation.

Part D

35. (a) $3(4) - 2(1) = 10$ (b) $5(-2) - (-1)(10) = 0$ (c) $2(3 - (-4)) - 0 + 1(2 - 12) = 14 - 10 = 4$
36. $A^{-1} = \frac{1}{10}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$
37. $AA^{-1} = \frac{1}{10}\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix} = \frac{1}{10}\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = I$. ✓
38. $\det = 0$ means the two rows are proportional (linearly dependent) — they describe the same line through the origin. No unique inverse exists.
39. $\det A = 6 - 5 = 1$; $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$; $\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}$
40. Near-zero determinant means tiny errors in $\mathbf{b}$ produce huge errors in $A^{-1}\mathbf{b}$, making the model numerically unstable.
41. $D = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$; $\det D = 6$. The determinant gives the factor by which areas are scaled.
42. $\det(AB) = 5 \times 3 = 15$.
43. (a) $4 + 7 = 11$ (b) $1 + 5 + 2 = 8$
44. $\lambda_1 + \lambda_2 = 5$, $\lambda_1\lambda_2 = 6 \Rightarrow (\lambda - 2)(\lambda - 3) = 0 \Rightarrow \lambda_1 = 2$, $\lambda_2 = 3$.

Part E

45. (a) 850 (b) 25 (c) $(850 + 920 + 970)/2900 \approx 0.945$, i.e. 94.5%
46. (a) $\sigma_1 = 2$, $\sigma_2 = 3$ (b) Yes; $\sigma_{ij} = \sigma_{ji}$ always — symmetry is a mathematical property of covariance.
47. $e^2 + e^1 + e^{0.5} \approx 11.76$; softmax $\approx [0.628, 0.231, 0.140]^T$
48. (a) Valid (b) Invalid (negative entry) (c) Valid (d) Invalid (sum $= 1.1 \neq 1$)
49. e.g. $\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ or $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

50. (a) $A\mathbf{x} = \mathbf{b}$ is the general form of a simultaneous system. (b) Dot products appear in every element of AB. (c) With millions of weights, the gradient is a vector; updating all weights at once is $\mathbf{w} \leftarrow \mathbf{w} - \alpha\nabla L$, a vector subtraction.